

Cross-Subject Sleep Stage Classification from EEG Signals Using Machine Learning and Deep Learning Models

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Abstract: Sleep stage classification is an essential part of diagnosing sleep disorders and understanding overall sleep health. Traditionally, sleep stages are identified by experts using Electroencephalography (EEG) recordings. However, this process takes a lot of time, requires significant effort, and the results may differ between clinicians. To overcome these challenges, this study explores the use of machine learning and deep learning methods to automatically classify sleep stages. Three models are implemented and compared: Random Forest, One-Dimensional Convolutional Neural Network (1D-CNN), and a hybrid CNN-LSTM model. The experiments are conducted on the publicly available Sleep-EDF Expanded dataset, with EEG signals undergoing preprocessing steps including band-pass filtering, Z-score normalisation, and segmentation into 30-second epochs. To compare the models fairly, Leave-Subject-Out (LSO) cross-validation is used so that the testing data contain only unseen subjects. Model performance is measured using Accuracy, Balanced Accuracy, Macro F1-score, Random Forest and Confusion Matrices. The experimental results indicate that the 1D-CNN achieves the best overall performance with an accuracy of 77.8%, while the CNN-LSTM attains an accuracy of 71.05%. The Random Forest classifier provides competitive results using handcrafted features but is less effective in learning the complex temporal patterns presents in EEG signals. Overall, the results show that lightweight deep learning models, especially the 1D-CNN, perform well., offer a good balance between classification performance and computational efficiency, making them suitable for cross-subject sleep stage classification in practical and resource-limited environments.

Index Terms — EEG, Sleep Stage Classification, Random Forest, 1D-CNN, CNN-LSTM, Deep Learning, Machine Learning, Sleep-EDF, Cross-Subject Generalisation

I. INTRODUCTION

Sleep is an important biological process that helps the body and brain recover, supports memory, and maintains overall health. Poor sleep or sleep disorders such as insomnia, sleep apnea, and narcolepsy can have serious effects on a person's physical and mental well-being. To diagnose these conditions, doctors use Polysomnography (PSG), a sleep study that records different physiological signals, including Electroencephalography (EEG),

which measures the brain's electrical activity during sleep.

Traditionally, sleep stages are identified manually by sleep specialists, who examine EEG recordings and classify every 30-second segment into one of five stages: Wake (W), N1, N2, N3, and Rapid Eye Movement (REM). Although this method is considered reliable, it is time-consuming, requires experienced professionals, and may produce different results depending on the scorer. As more sleep studies are conducted, there is a greater need for accurate and automated sleep stage classification systems.

Recent advances in machine learning and deep learning have made it possible to classify sleep stages automatically using EEG signals. However, many existing studies evaluate their models' using data from the same subjects for both training and testing. This can lead to overly positive results because EEG signals vary from person to person. In real-world clinical applications, models should be able to classify sleep stages for people they have never seen before.

This project focuses on cross-subject sleep stage classification, where the training and testing data come from different individuals. This approach provides a more realistic evaluation of a model's ability to generalise to new subjects. The study uses the Sleep-EDF Expanded dataset and compares the performance of three different models: Random Forest, One-Dimensional Convolutional Neural Network (1D-CNN), and CNN-LSTM. The models are evaluated using performance metrics such as Accuracy, Balanced Accuracy, Macro F1-score and Confusion Matrices. By comparing these approaches, this study aims to identify a model that provides accurate sleep stage classification while remaining efficient enough for implementation in environments such as Google Colab.

II. PROBLEM STATEMENT

Sleep stage classification is an important part of diagnosing sleep disorders and monitoring a person's sleep quality. Traditionally, sleep stages are identified manually by sleep experts using EEG recordings collected during overnight sleep studies. Although this method provides reliable results, it is slow, time-consuming, and requires experienced clinicians. As the number of sleep studies continues to increase, there is a

growing need for automated systems that can classify sleep stages accurately and efficiently.

Many machine learning and deep learning models have been developed for automatic sleep stage classification. However, most studies evaluate these models using data where recordings from the same individuals are used in both the training and testing sets. This type of evaluation often produces high accuracy because the model has already learned the EEG patterns of those individuals. In real clinical applications, the model must classify sleep stages for new patients whose EEG signals were not included during training. Since EEG signals vary from person to person due to differences in brain activity, age, health conditions, and recording conditions, the model's performance often decreases when tested on unseen subjects. This makes cross-subject generalisation one of the biggest challenges in automated sleep stage classification.

Another challenge is the imbalance in sleep stage data. Some sleep stages, particularly N1, occur much less frequently than stages such as N2 or Wake. Because of this imbalance, classification models tend to predict the majority classes more accurately while making more errors on the minority classes. This reduces the overall reliability of the system, especially for clinically important sleep stages.

To address these challenges, this project compares three different classification models: Random Forest, One-Dimensional Convolutional Neural Network (1D-CNN), and CNN-LSTM using the Sleep-EDF Expanded dataset. EEG signals are preprocessed and evaluated using the Leave-Subject-Out (LSO) validation method, where the training and testing data come from different subjects. This evaluation provides a more realistic measure of how well each model performs on unseen individuals. The models are compared using Accuracy, Balanced Accuracy, Macro F1-score and Confusion Matrices. The main goal of this study is to develop a model that accurately classifies sleep stages, uses fewer computing resources, and can be easily implemented on Google Colab and in future healthcare applications.

III. OBJECTIVES

The main objectives of this project are:

- To develop an automated system for classifying sleep stages using EEG signals from the Sleep-EDF Expanded dataset.
- To implement and compare three different classification models: Random Forest, 1D Convolutional Neural Network (1D-CNN), and CNN-LSTM.
- To evaluate the performance of these models using Leave-Subject-Out (LSO) cross-validation, where the training and testing data come from different subjects.
- To measure and compare the performance of each model using evaluation metrics such as Accuracy,

Balanced Accuracy, Macro F1-score and Confusion Matrices.

- To analyse how well each model classifies the five sleep stages: Wake (W), N1, N2, N3, and REM.
- To identify the model that provides the best balance between classification accuracy, generalisation to unseen subjects, and computational efficiency.
- To develop a lightweight and practical sleep stage classification system that can be implemented on Google Colab and support future healthcare and sleep monitoring applications.

IV. DATASET

We have chosen Sleep-EDF Expanded Database, which is publicly available through PhysioNet [2], and this dataset only we used in this study to examine all results and get perfect output. This is heavy helpful which we get dataset from physionet and its shape our project study. We chose dataset that having overnight polysomnography recordings collected from healthy volunteers and individuals with mild sleep disorders or some normal persons which having good sleep. What we choose dataset that Recordings having EEG, Electrooculography (EOG), and Electromyography (EMG) signals / Waves, along with expert-annotated sleep stage labels.

For this study, we have used only the EEG channel Fpz-Cz, which provides reliable frontal brain activity relevant to sleep staging over the night. We have recordings and which are separated into 30-second epochs, also we have labelled to each according to the Rechtschaffen and Kales (R&K) standard into following of five stages: Wake (W), N1, N2, N3, and REM. The dataset exhibits inherent class imbalance, with N2 being the most frequent stage and N1 the least, which affecting model training and evaluation under cross-subject protocols but this how we are getting better results after examining.

Below are the URL for dataset we have used during our project: <https://physionet.org/content/sleep-edfx/>

V. METHODOLOGY

Below are details methodology we used during our project we highlighted each and every part very carefully and perfectly, the explanation are as follows.

A. Data Preprocessing:

The preprocessing pipeline we used, we verified data examine and later cleaned (normalized) data and then applied consistently across all three models to ensure fair comparison.

Below are the steps we followed during our projects:

- Loading raw EEG recordings and check and verified from the Sleep-EDF Expanded dataset.
- After verification carefully selecting the Fpz-Cz EEG channel for all recordings.

- Adding, removing (normalized) artefact-contaminated and noisy segments (outliers) prior to analysis.
- We applied the 4th-order Butterworth band pass filter with a passband of 0.3–35 Hz to remove baseline drift and high-frequency noise so we can get cleared data.
- Then we did down sampling signals to 100 Hz to reduce computational load while preserving diagnostically relevant frequency content and coz of this we got more clarity in models.
- Then after we did segment the continuous EEG into non-overlapping 30-second epochs and continuous, consistent with clinical scoring conventions.
- End of the we applied Z-score normalisation to each epoch to standardise amplitude across subjects and reduce inter-subject variability to get more clarity.

B. Feature Extraction — We Used Random Forest (Model-1):

Structured numerical input features are required for Random Forest, instead of raw EEG signals. Hence, from each EEG epoch, the spectral features had been extracted.

To capture the characteristics of EEG signal that corresponds to different stages of sleep, analysis on frequency-domain and time-domain were performed. The Time-Domain includes the following statistics: mean, standard deviation, maximum, minimum, variance and the peak-to-peak amplitude, which provides the summary of the shape and waveform's variability. The features of Frequency-Domain were extracted by computing the spectral density applying Welch's method and the band power was extracted for the four frequency bands: Delta (0.5-4 Hz), That (4-8 Hz), Alpha (8-13 Hz), and Beta (13-30 Hz). These ten features, which are the six time-domain statistics and the four frequency-domain band powers, had been computed individually for each of two EEG channels: Fpz-Cz and Pz-Oz. This gives a vector of 20-dimensional feature. The combination of the frequency-domain and time-domain, gives informative depiction for differentiating the sleep between Wake, NREM and REM.

These features provide an outline on the distribution of energy across the different frequency, which provides information that are useful to differentiate between the sleep: Wake, NREM, and REM sleep.

The feature vectors that are extracted were combined to the structural dataset, in which each row describes one EEG epoch and each column describes the numerical feature. These were used as the inputs to Random Forest Classifier.

This approach gives a strong baseline as it depends on domain knowledge than the automatic learning of feature.

Below are some details, features are extracted from after used this model:

- Power Spectral Density (PSD) consider values across the full frequency range.
- Frequency band power: Delta (0.5–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz) this all consideration we did.
- Following Statistical features we observed: Mean, Standard Deviation, Variance.
- Following are Hjorth parameters: Activity, Mobility, and Complexity.
- Then later we used Spectral Entropy to quantify EEG signal irregularity.

Random Forest was chosen as baseline ML model, due to its simplicity, robustness and the capability to handle or manage the nonlinear relationship between the features.

Multiple decision trees were constructed using the bootstrap sampling on the training data. Each of the decision tree predicts the sleep stage independently, and majority voting across all the trees, determines the final prediction. This strategy enhances the stability of the prediction while decreasing the risk or chance of overfitting.

Random Forest provides numerous advantages and benefits for the biomedical datasets. It performs or executes well on the structured features, involves somewhat little hyperparameter tuning, and gives steady performance although when some features contain some noise.

The Random Forest model was trained on the spectral features which were extracted and were evaluated using the validation called Leave-One-Subject-Out to establish as the classical ML baseline.

C. One-Dimensional Convolutional Neural Network (1D CNN) (Model-2)

Unlike the Random Forest, the 1D CNN had been designed to learn the features directly from the raw EEG signals without the need for feature engineering manually.

The network contains numerous one-dimensional convolutional layers that learned the local temporal patterns automatically from EEG signals. On each convolution layer, the activation function and pooling operations were performed that progressively transformed the raw waveform into representation of high-level features. To reduce the problem of overfitting, dropout layers were integrated which disables the neurons randomly during the training. After the feature extraction, the fully connected layers were utilized to implement final classification of the sleep stage by a softmax output layer. The CNN enhances the model and enables to discover and learn the characteristics of signals that are meaningful, which the handcrafted spectral features may not capture, thus making it suitable especially for the classification of EEG signals.

Below are some details, features are extracted from after used this model:

- We have used Input Layer (3000 samples per epoch at $100 \text{ Hz} \times 30\text{s}$)
- Then Conv1D $\rightarrow$ Batch Normalisation $\rightarrow$ ReLU Activation
- Then MaxPooling1D
- Then Conv1D $\rightarrow$ Batch Normalisation $\rightarrow$ ReLU Activation
- Then Global Average Pooling
- After all these above points we got Softmax Output Layer (5 classes)

When we used Global Average Pooling then its reduced overfitting after minimising the number of trainable parameters, and which makes the model well-suited for cross-subject generalisation and we got good results.

D. CNN-LSTM Hybrid Model (Model-3)

CNN-LSTM was implemented to further enhance the performance. Local spatial and the temporal features were extracted from EEG signals. These feature maps were fed to the LSTM (Long Short-Term Memory) layer.

LSTM networks are specifically very effective for data that are sequential, as they maintain and contains the previous information's memory through a specific gating mechanism. This allows the network to obtain the temporal relationships that may go beyond the local receptive field. The hybrid architecture of combining the extraction from the CNN feature with the LSTM sequence benefits from automatic representation learning and modelling of temporal dependency. This makes the model suitable for EEG signal analysis, in which both the characteristics of local waveform and temporal dynamics, which then contributes to the accurate classification of the sleep stages.

Below are some details, features are extracted from after used this model:

- First layer is Input Layer
- Then secondly Conv1D $\rightarrow$ MaxPooling1D
- Then Conv1D $\rightarrow$ MaxPooling1D
- And then LSTM Layer (64 units)
- end of the Dropout (rate = 0.3)
- Then Moved to Dense Layer (64 units, ReLU)
- End of the model we used Softmax Output Layer (5 classes)

This model LSTM layer enabled and used the model to retain contextual information across time steps within each 30-second epoch and so on, and that is things which is theoretically beneficial for capturing the gradual transitions between sleep stages all over the night.

Leave-One-Subject-Out Cross-Validation

Leave-One-Subject-Out (LOSO) was done to evaluate the capability of generalization for all the three models.

In each of the iteration, all the EEG epochs that belongs to one subject were exclusively reserved for testing, whereas the training dataset were formed from the remaining subjects. This process was repeated until every subject had been served once as part of the test subject. Unlike the random train-test split, LOSO helps to prevent the subject information from arriving in both the training and testing, therefore it eliminates the subject leakage. When this evaluation protocol is applied on completely new subjects, it provides a realistic assessment on the performance.

VI. EXPERIMENTAL SETUP

Mostly we are working on Google colab to build models train it and for results, we are avoiding local resources to use. Code we are doing in python UI we used google colab also we are using cloud computing environment to ensure reproducibility without requiring local GPU resources. For data we used google drive we mapped drive to colab and we fetched data directly from google drive.

Below are some details, libraries are used in this model:

The following are used:

- NumPy and Pandas — data loading and manipulation.
- MNE — EEG signal processing and epoch extraction.
- SciPy — signal filtering and spectral feature computation.
- Scikit-learn — Random Forest training and evaluation metrics.
- TensorFlow and Keras — deep learning model construction and training.
- Matplotlib — visualisation of results and confusion matrices.

Evaluation Strategy: Leave-Subject-Out (LSO) cross-validation is applied throughout so it goes smoothly. In each and every fold, we consider one subject is held out entirely as the test set, and the remaining subjects are used for training so we will get better results after that. This process is repeated until every subject has served as the test subject exactly once and remaining for test. This technic strictly prevents data leakage between training and testing so we will get realistic estimate of model generalisation to unseen individuals and the best results.

VII. PERFORMANCE METRICS

Below are the details which evaluation metrics are we used to assess and compare model performance and get best results:

- Accuracy: We used the proportion of correctly classified epochs in all five sleep stages to get best results.
- Balanced Accuracy: We used the average per-class recall, accounting for class imbalance in the dataset to get best results.

- **Macro F1-Score:** We used the unweighted mean of per-class F1-scores, treating each and every sleep stages equally regardless of frequency to get best results.
- **Cohen's Kappa:** We used a measure of inter-rater agreement between the model's predictions and ground truth labels, correcting for chance agreement and to get best results.
- **Confusion Matrix:** We used a detailed breakdown of correct and incorrect classifications for each sleep stage pair, enabling analysis of specific misclassification patterns to get best results.

The This here very important are Balanced Accuracy and Macro F1-Score are somewhat important in this context due to the significant class imbalance in the dataset, and the minority stages for example as N1 are substantially underrepresented.

VIII. EXPERIMENTAL RESULTS

A. Model Accuracy Comparison

The Table I represents the accuracy that were achieved by each of the model under the LOSO cross-validation.

Model	Accuracy (%)
Random Forest	78.54
1D CNN	77.80
CNN-LSTM	71.05

B. Confusion Matrix Analysis

For all of the three model confusion matrix was generated to assess and analyze the classification behavior.

Random Forest: The strongest per-class F1 score is achieved in Wake, N2 and N3 epochs with high reliability. N1 is the weakest stage, in which the N1 can be misclassified into N2 and REM which is common.

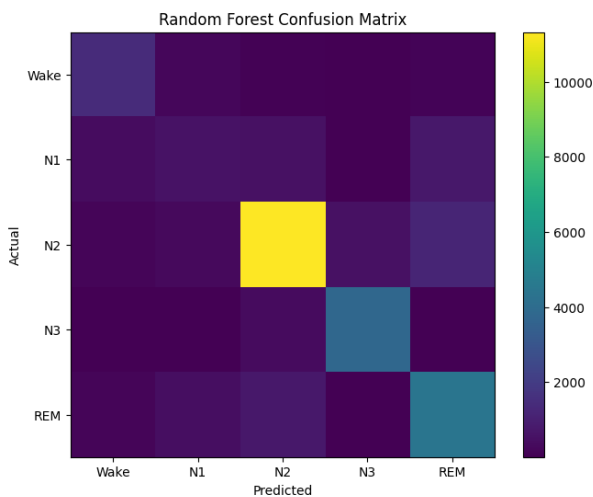


Figure 1. Represents the Confusion Matrix of Random Forest

The representational capacity is limited to models, since the random forest relies mainly on the 20-dimensional

handcrafted vector, compared to the models that can learn the features directly from the raw signal – however this does not prevent the RF from achieving the best Macro F1 Score overall.

1D CNN: The most balanced confusion matrix was produced by 1D CNN across all the five sleep stages. This was due to that the model can automatically learn the features from the raw waveform; therefore, it allows to identify the patterns that are task related, without the need of relying on the domain-specific feature engineering.

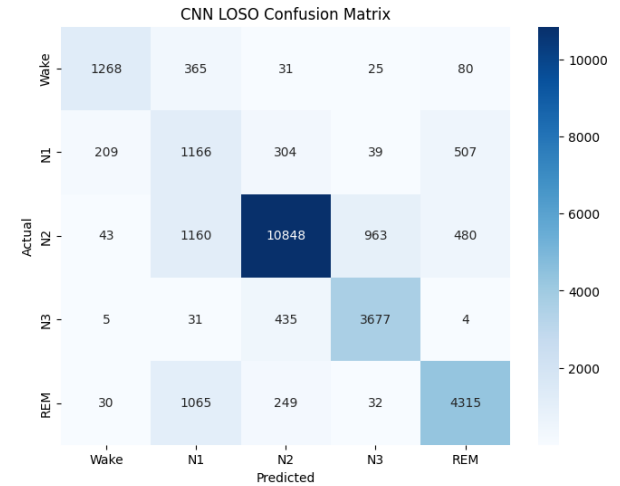


Figure 2. Represents the Confusion Matrix of 1D-CNN

CNN-LSTM: The CNN-LSTM was intended to grasp the sequential dependencies between epochs, but this model had underperformed the 1D CNN on every metric. It misclassifies N1 and N3 epochs, and the confusion matrix had shown more errors across the N2, N3 and REM.

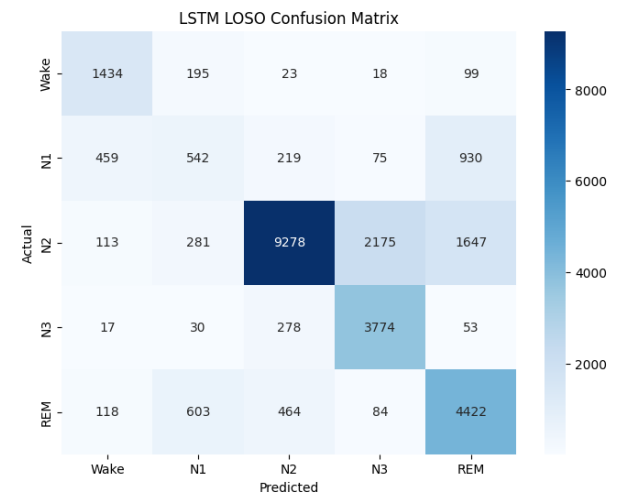


Figure 3. Represents the Confusion Matrix of CNN LSTM

REM is occasionally confused with Wake on across all three models – which is consistent by using only the EEG as input, as the REM from EOG and EMG were excluded and not available to this model.

IX. DISCUSSION

There is a mixed picture of results than a single clear winner. The highest Macro F1 (0.6996) was achieved by Random Forest, the metric which is most robust to the imbalance in the class in dataset. The highest balanced accuracy (0.7300) and Cohen's Kappa (0.6779). This indicates that domain-specific spectral and statistical features from the well-engineered baseline have been competitive with the Macro F1 which is slightly ahead of the deep learning model. The CNN-LSTM has achieved the lowest score on all metrics. The Random Forest and 1D CNN are trained with the same LOSO cross-validation, hence we cannot simply state "insufficient data under LOSO". Overall, these results help to form two conclusions: choice of model between classical and deep learning depends on which of the metric is essential and matters most. And the other conclusion, the complexity of the added architecture is not beneficial automatically without training budget sufficient to support.

Overall, these findings support the use of compact convolutional architectures for cross-subject sleep stage classification tasks, particularly in resource-constrained deployment environments.

X. LIMITATIONS

This Project relies only on EEG signals (Fpz-Cz and Pz-Oz), despite the dataset contains additional physiological signals. EOG and EMG were excluded; these signals are the standard clinical discriminators for the REM sleep. This may likely explain the confusion of Wake/REM. The imbalance in the class N1 stage affects the model training and it's reflected across all three models in the low F1 scores. Restriction on the training data volume per fold due to the limited subject pool of 30 subjects which were used in LOSO cross-validation.

Also, we face data imbalanced and not cleared. Accessibility issue because data was huge.

XI. FUTURE WORK

We found some future work here we have mention below here:

- As per our study we found Transformer-based architectures also we found attention mechanisms for improved long-range EEG dependency modelling.
- Secondly, we found Multi-channel EEG analysis incorporating EOG and EMG signals to replicate full polysomnography input which also we can consider in future.
- Transfer learning approaches also we can used to reduce the impact of inter-subject variability and also improve model personalisation this option also good.
- Amin thing Data augmentation strategies for finding class imbalance, particularly for the underrepresented N1 stage.

- Federated learning frameworks to enable multi-site model training while preserving patient data privacy.
- Real-time deployment on wearable EEG devices for continuous ambulatory sleep monitoring.

XII. CONCLUSION

This project provides a comparative analysis of three models – Random Forest, 1D CNN, and CNN_LSTM using 30 subjects for cross-subject sleep stage classification. All three models were evaluated and validated using Leave-One-Subject-Out (LOSO) cross-validation, using Accuracy, Balanced Accuracy and confusion matrix.

The highest Macro F1 Score was achieved by Random Forest, whereas 1D-CNN achieved highest balanced accuracy and Cohen's Kappa. Lowest scores were achieved by CNN-LSTM, might be because of under training within a fixed fold. The most difficult sleep stage to classify was N1 on all three models, which is also consistent with known clinical ambiguity.

These findings and observations show that leakage-free cross-subject validation and domain-informed feature engineering matters as much as the architecture of the model for automated sleep staging, and that neither deep learning nor classical machine learning should be assumed to dominate the other without protocol-control, metric-specific evaluation.

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